

K-FIELD

DUAL-AXIS KINETIC MASS CMB TRAVERSAL GEOMETRY

Earth-center 2:1 compound rotation translated through the ICRS-embedded K-Field
along the Earth-CMB vector, with an RA=0 deg seasonal datum

Earth-center dual-axis traversal through the ICRS-embedded K-Field

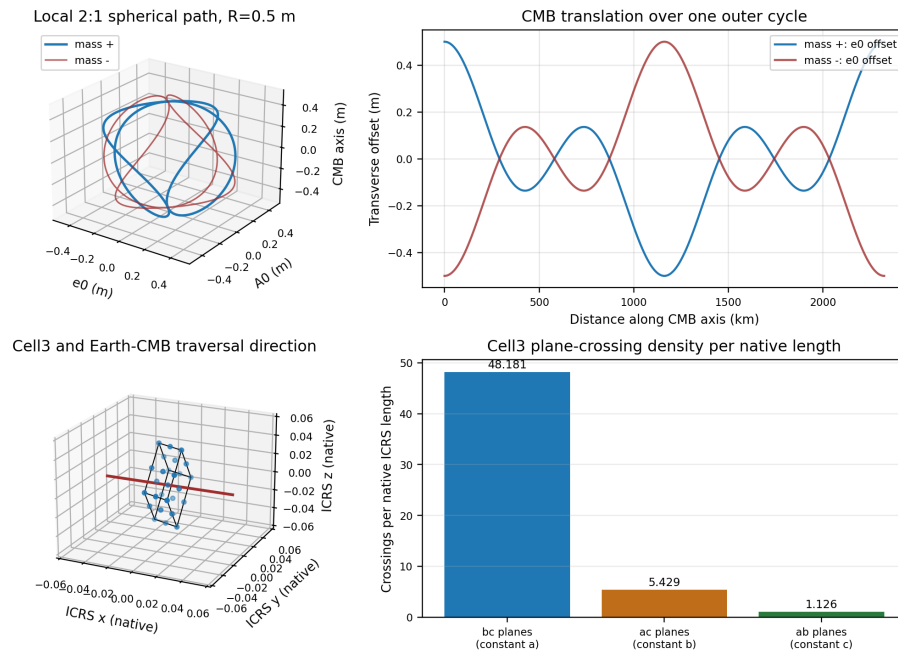


Figure 1. Title composite. Four computed views of the same construction: the local spherical endpoint curve, the long-baseline CMB translation, the Cell3 base geometry, and the exact plane-crossing density.

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Archival computational edition - Unix epoch 1784745791

Frontispiece: authoritative Cell3 base cell, face planes, and all nodes

Authoritative Cell3 base cell in ICRS: six faces, 12 edges, and all 27 nodes

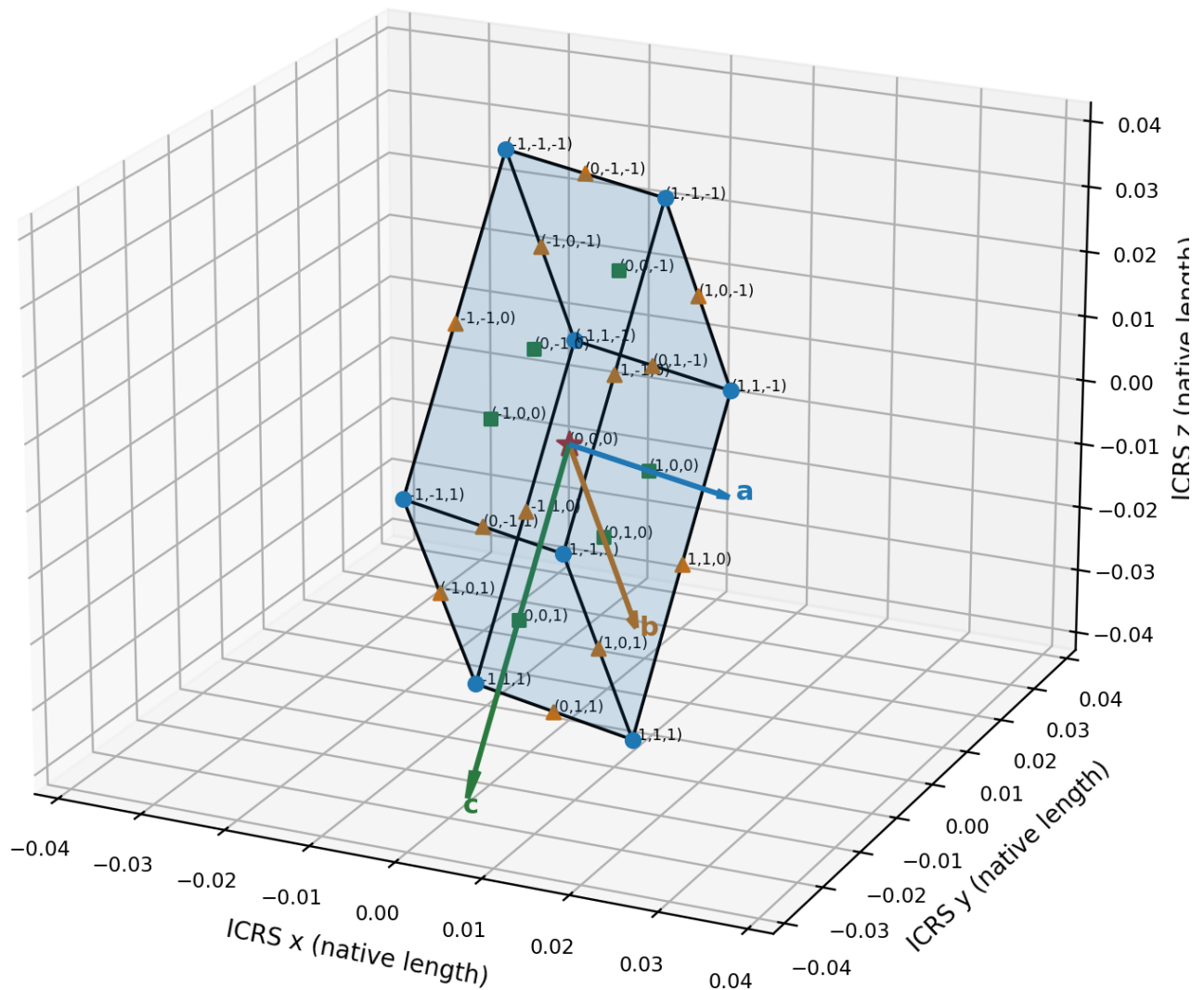


Figure 2. Authoritative Cell3 in ICRS. The centered primitive Cell3 is rendered from the source basis. All six boundary faces, 12 edges, 27 centered half-step nodes, integer node labels, and basis vectors are shown in native dimensions.

Abstract

This reference defines the geometric path produced when the exact 2:1 dual-axis endpoint motion is placed at Earth center and translated through the ICRS-embedded K-Field along the Earth-CMB vector. The local apparatus consists of two antipodal point masses at radius $R=0.5$ m. Axis B is fixed to the Earth-CMB direction, axis A remains perpendicular to axis B, and the inner angular rate is twice the outer angular rate. In the Earth-comoving frame, each endpoint traces a closed spherical algebraic curve with first-, second-, and third-harmonic components. In the fixed K-Field frame, Earth translation converts that closed curve into an open, translation-periodic two-strand path contained inside a radius-R tube around the Earth-CMB centerline.

An idealized seasonal datum is selected at Earth-center right ascension 0 deg and declination 0 deg, one astronomical unit along the positive ICRS x direction. This datum fixes the initial location without adding a terrestrial latitude, diurnal rotation, or a calendar ephemeris. The CMB translation remains the declared motion channel. At the unit outer rate $\omega_B=1$ rad/s, one outer cycle lasts 2π seconds and advances the Earth center 2,323.600478980 km along the CMB axis. The 0.5 m local modulation is therefore a fine structured weave around a nearly straight global centerline.

The same path is transformed into Cell3 coefficient coordinates. The Earth-CMB line has direction $B_K^{-1} \hat{c}=(-48.18103814, 5.42930495, 1.12575307)$. It is oblique to every Cell3 edge and face family. Per native ICRS length it crosses the constant-a or bc plane family at density 48.18103814, the constant-b or ac family at density 5.42930495, and the constant-c or ab family at density 1.12575307. The physical spacing requires a separate scale bridge ℓ_K between one native K-Field length and meters; the source defines the geometry and orientation but does not supply that bridge.

Central result. The resulting path is a translationally periodic, antipodal, four-lobed compound braid advancing along the Earth-CMB vector and cutting obliquely through all three Cell3 plane stacks. The geometry separates into a global directed translation, a bounded spherical modulation, and a discrete plane-crossing address sequence. This decomposition supplies a direct pathway from mechanical phase to K-Field position, plane family, local direction, and repeated traversal phase.

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1. Purpose, scope, and geometric hierarchy

The objective is to describe one mechanical motion in three compatible geometric layers. The first layer is the local endpoint curve measured from Earth center. The second is the absolute traversal obtained by translating Earth center along the Earth-CMB velocity vector. The third is the representation of that absolute traversal in the oblique Cell3 coordinate system. Keeping these layers distinct prevents local rotor coordinates, ICRS coordinates, and Cell3 coefficients from being conflated.

The device is placed at Earth center. This removes surface latitude, longitude, diurnal rotation, and local vertical from the construction. Earth center is the common origin of the mass pair at the selected datum. The only imposed global translation is the Earth-CMB motion used by the source analysis. The seasonal datum fixes an initial ICRS position and does not change the local dual-axis equations.

The construction uses the source report and machine-readable dataset identified on the title page. The numerical constants, basis vectors, node coordinates, plane equations, Q matrix, and endpoint radius are reproduced directly from those files. Derived quantities in this reference are computed from those values without altering their orientation or normalization.

Quantity	Definition	Value
Mass pair	Two equal endpoint point masses	1 kg each
Bar	Rigid massless center-to-center constraint	1 m total length
Endpoint radius	Distance from Earth-center hub to each mass	0.5 m
Axis B	Outer axis and translation direction	Earth-CMB unit vector
Axis A	Inner axis	Perpendicular to axis B
Rate lock	Inner to outer angular rate	$\omega_A / \omega_B = 2$
Unit-rate witness	Outer and inner rates	1 rad/s and 2 rad/s
Seasonal datum	Earth-center ICRS position	RA=0 deg, Dec=0 deg, radius=1 AU

2. Coordinate frames and authoritative vectors

2.1 ICRS Cartesian frame

The International Celestial Reference System is used as a fixed Cartesian orientation. The positive x axis points to RA=0 deg, Dec=0 deg; the positive y axis points to RA=90 deg, Dec=0 deg; and the positive z axis points to the north celestial pole. Every vector in this reference is expressed first in that frame.

2.2 Earth-CMB axis and local path frame

Object	ICRS x	ICRS y	ICRS z	Magnitude
Earth-CMB axis	-0.970770855836	0.207348009111	-0.120874929482	1
Earth-CMB velocity (m/s)	-359003.199066	76679.886034	-44701.060101	369812.502000
Bar reference e0	-0.200864884110	-0.426231479950	0.882031758970	1
Axis A reference A0	0.131366829113	0.880530254232	0.455422032396	1

The ordered triad (e0, A0, c_hat) is orthonormal and right-handed. It is a moving-mechanism frame embedded in fixed ICRS space. The local coordinates X, Y, and Z used below are projections onto e0, A0, and c_hat, respectively.

Idealized seasonal datum and path frame in ICRS
 Angle from RA=0 datum to CMB axis = 166.112975 deg

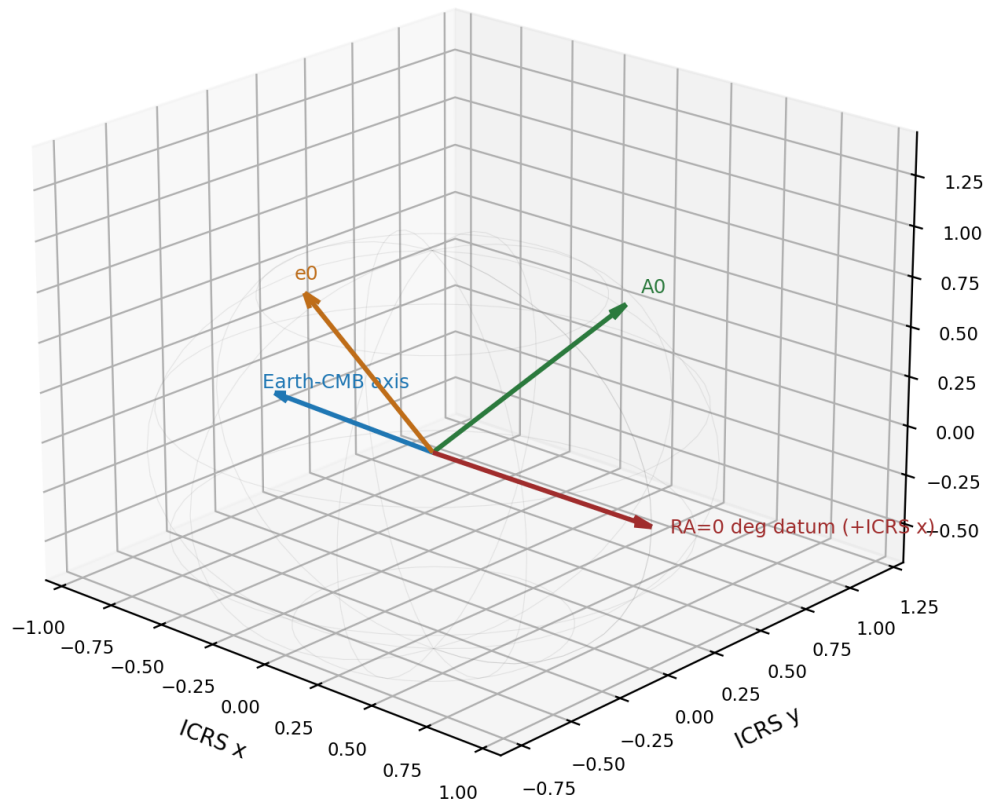


Figure 3. Seasonal datum and path frame in ICRS. The idealized Earth-center datum lies on +ICRS x. The Earth-CMB axis is 166.112975 deg from that datum direction. The transverse vectors e0 and A0 complete the right-handed path frame.

3. The exact local 2:1 compound-rotation curve

3.1 Mechanical phase construction

Let $\phi = \omega_B t$ be the phase of the outer rotation about the Earth-CMB axis. The transverse unit vector $e(\phi)$ rotates once about $\hat{c}$ during one outer cycle. The bar simultaneously turns through inner phase 2ϕ . The positive endpoint direction $u(\phi)$ is therefore formed by combining the rotating transverse direction with the fixed CMB-axis direction.

$$e(\phi) = \cos(\phi) e_0 + \sin(\phi) A_0$$

$$u(\phi) = \cos(2\phi) e(\phi) + \sin(2\phi) \hat{c}$$

$$r_{\text{plus}, \text{local}}(\phi) = +R u(\phi); r_{\text{minus}, \text{local}}(\phi) = -R u(\phi)$$

Because $e(\phi)$ is perpendicular to $\hat{c}$ and both are unit vectors, $u(\phi)$ is a unit vector for every phase. Each endpoint remains exactly $R=0.5$ m from Earth center. The pair remains antipodal and the bar length remains exactly 1 m.

3.2 Fixed-frame harmonic expansion

$$u(\phi) = 0.5[\cos(3\phi) + \cos(\phi)] e_0 + 0.5[\sin(3\phi) - \sin(\phi)] A_0 + \sin(2\phi) \hat{c}$$

This expression exposes the curve as a finite harmonic object. The e_0 and A_0 components contain the first and third harmonics; the CMB-axis component contains the second harmonic. The path closes after ϕ advances by 2π because every harmonic then returns to its starting value.

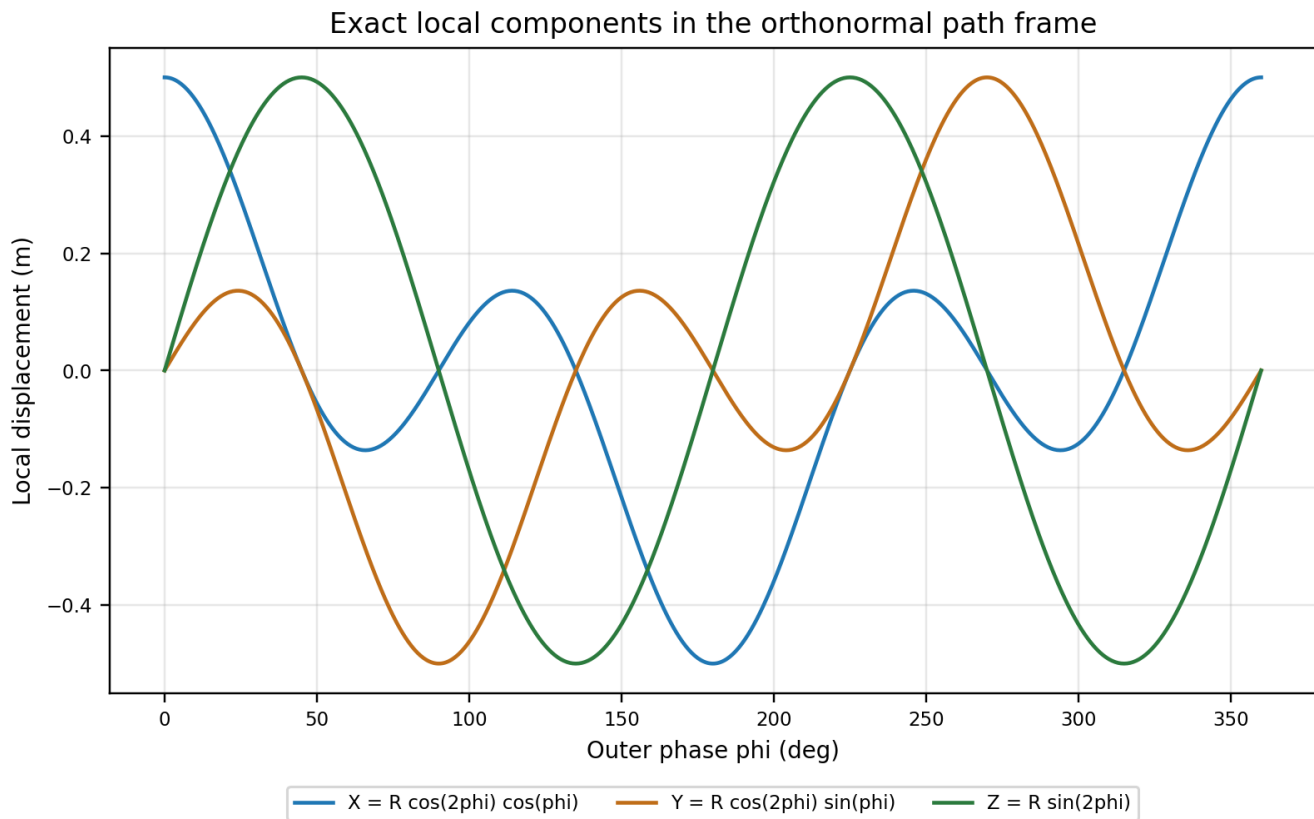


Figure 4. Local path components. The three exact displacement components are plotted over one outer cycle. The axial component completes two full oscillations while the transverse direction completes one outer revolution.

3.3 Intrinsic spherical and algebraic geometry

$$X = R \cos(2\phi) \cos(\phi); Y = R \cos(2\phi) \sin(\phi); Z = R \sin(2\phi)$$

$$X^2 + Y^2 + Z^2 = R^2$$

$$Z (X^2 + Y^2) = 2 R X Y$$

The first identity proves that the local path lies exactly on the sphere of radius R . The second eliminates the phase and identifies the path as the real intersection of that sphere with a cubic surface. It is therefore a closed immersed spherical algebraic curve rather than a planar figure-eight or an ordinary circle.

The curve has four pole-to-equator sectors. It reaches the positive CMB pole twice and the negative CMB pole twice during one cycle. The repeated pole visits are tangential self-contacts in the local geometric locus, not cusps: the parameter derivative never vanishes. The exact local curve length at $R=0.5$ m is 6.659167222 m.

Closed 2:1 compound-rotation curve in the Earth-comoving frame

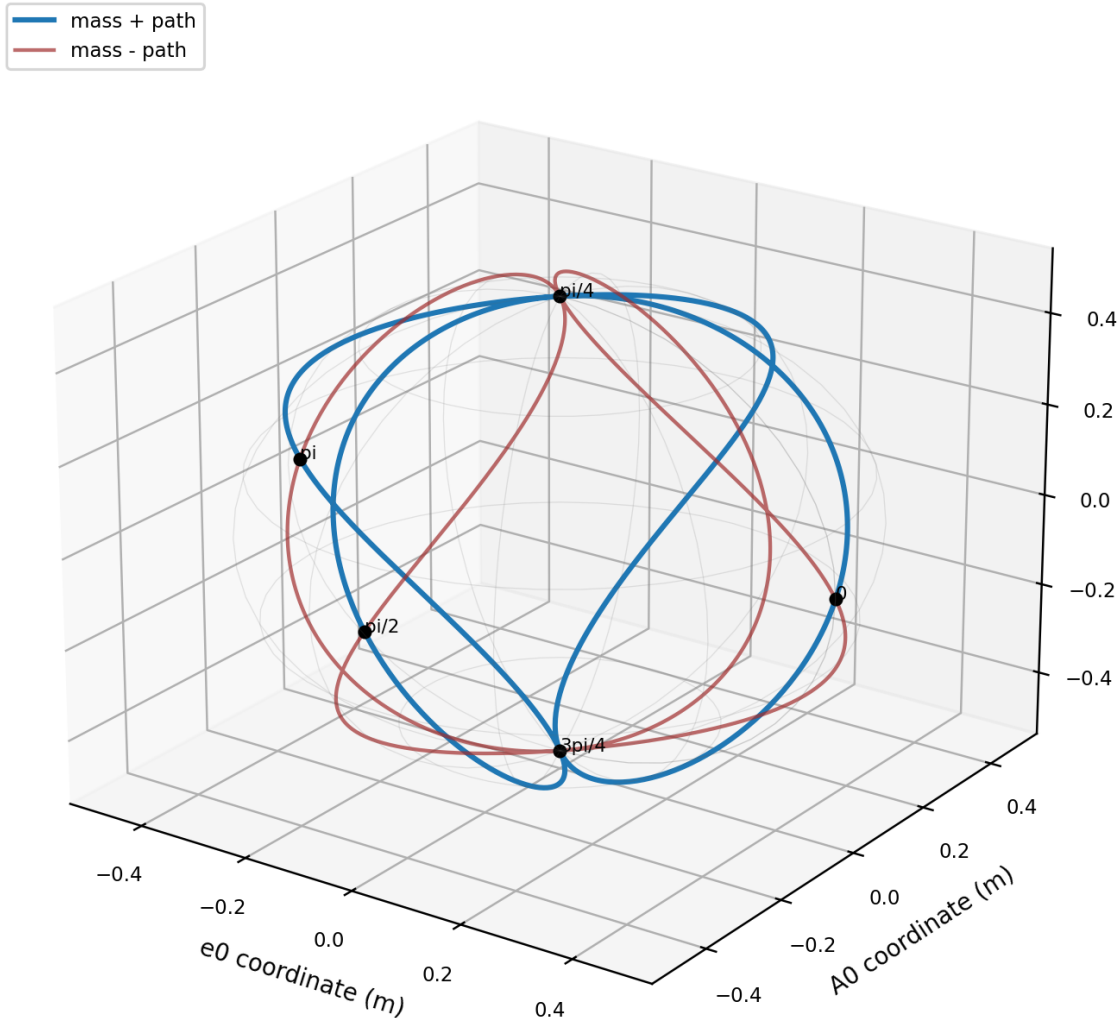


Figure 5. Local closed spherical path. The positive and negative endpoint paths are shown in the orthonormal path frame on the radius-0.5 m sphere. The two masses are pointwise antipodal.

4. Seasonal Earth-center datum at RA=0 deg

A seasonal datum is required only to anchor the absolute ICRS position of the Earth-center device. The chosen datum places Earth center one astronomical unit along the positive ICRS x axis:

$$r_{E0} = AU \hat{x}_{ICRS} = (149597870700, 0, 0) \text{ m}$$

This is an idealized heliocentric coordinate datum. It does not assign a calendar date, orbital eccentricity, or orbital velocity. Those quantities are unnecessary for the declared CMB-directed traversal. At the datum, the device center is exactly Earth center and the local mass offsets are added directly to r_{E0} .

The angle between the datum radius and the Earth-CMB axis is 166.112974512 deg. Since $\hat{c}$ has x component -0.970770855836, the CMB-directed translation points predominantly toward decreasing ICRS x from this datum.

The initial absolute endpoint positions at $\phi=0$ are:

$$r_{plus}(0) = r_{E0} + R \hat{e}_0$$

$$r_{minus}(0) = r_{E0} - R \hat{e}_0$$

5. Translation through the K-Field along the Earth-CMB vector

5.1 Earth-center trajectory

In the fixed K-Field frame, Earth center is not stationary. Under the declared CMB translation model, its position advances linearly along $\hat{c}$. Using ϕ as the independent variable gives:

$$r_C(\phi) = r_{E0} + (V_{CMB} / \omega_B) \phi \hat{c}$$

The two absolute endpoint paths are the translated Earth-center line plus the bounded local spherical offsets:

$$r_{plus}(\phi) = r_{E0} + (V_{CMB}/\omega_B) \phi \hat{c} + R u(\phi)$$

$$r_{minus}(\phi) = r_{E0} + (V_{CMB}/\omega_B) \phi \hat{c} - R u(\phi)$$

This formula is the complete geometric definition of the traversal. It contains no approximation in the local 2:1 motion. At unit outer rate, the only idealization is the use of constant Earth-CMB translation during the cycle.

5.2 Axial and transverse decomposition

$$s_{plus}(\phi) = (V_{CMB}/\omega_B) \phi + R \sin(2 \phi)$$

$$s_{minus}(\phi) = (V_{CMB}/\omega_B) \phi - R \sin(2 \phi)$$

$$\rho_{plus}(\phi) = +R \cos(2 \phi) [\cos(\phi)e_0 + \sin(\phi)A_0]$$

$$\rho_{minus}(\phi) = -R \cos(2 \phi) [\cos(\phi)e_0 + \sin(\phi)A_0]$$

$$|\rho(\phi)| = R |\cos(2 \phi)|$$

The path repeatedly contracts to the CMB centerline and expands back to radius R . The centerline encounters occur at $\phi=\pi/4$, $3\pi/4$, $5\pi/4$, and $7\pi/4$. At each encounter the two masses lie at different axial positions separated by $2R$, so the antipodal bar remains intact. The global translation separates repeated local pole visits into distinct points along the CMB line.

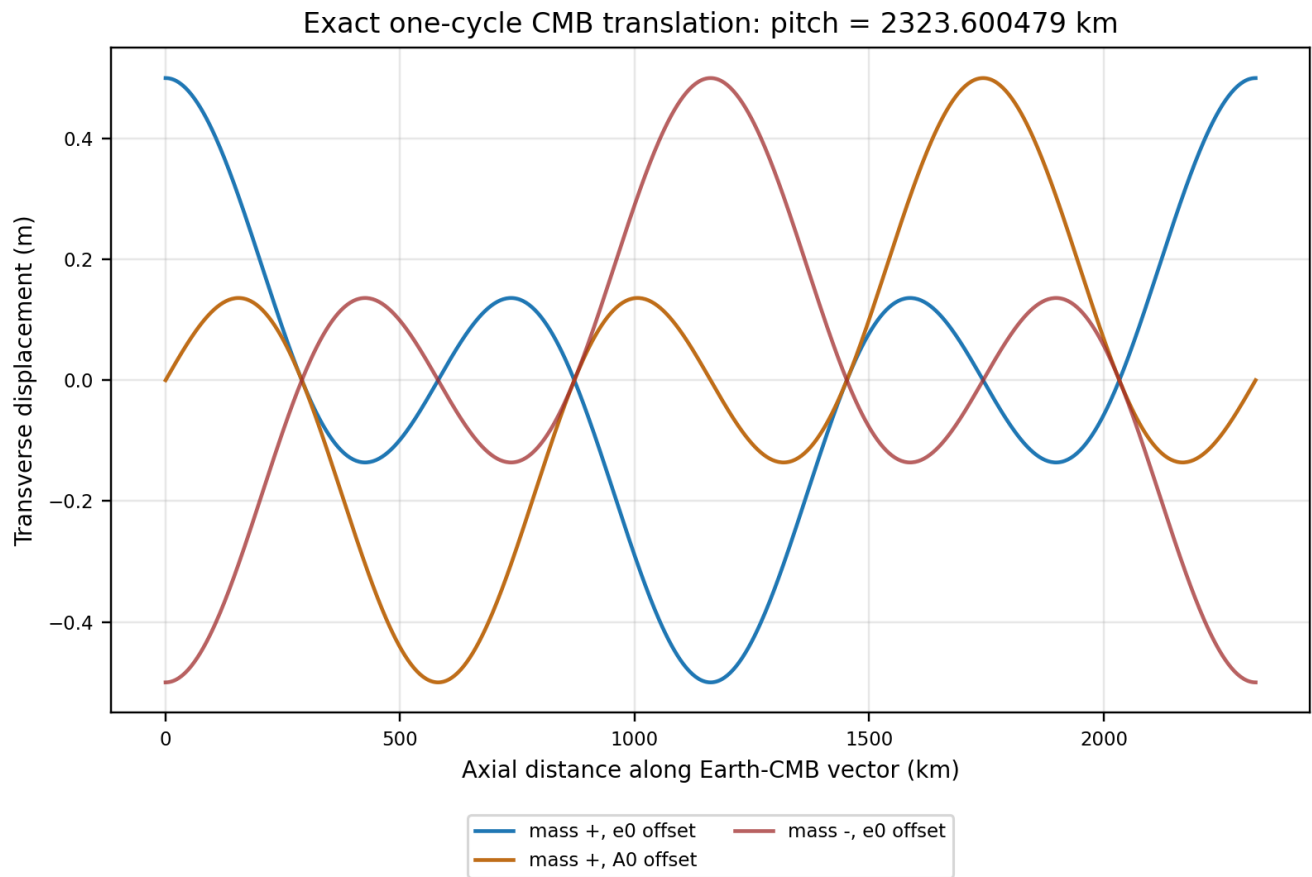


Figure 6. Exact global progression. The axial coordinate is plotted in kilometres and the transverse coordinates in metres. The different units expose the 0.5 m modulation without altering the computed physical values.

Dimensionless normalized two-strand traversal over one outer cycle

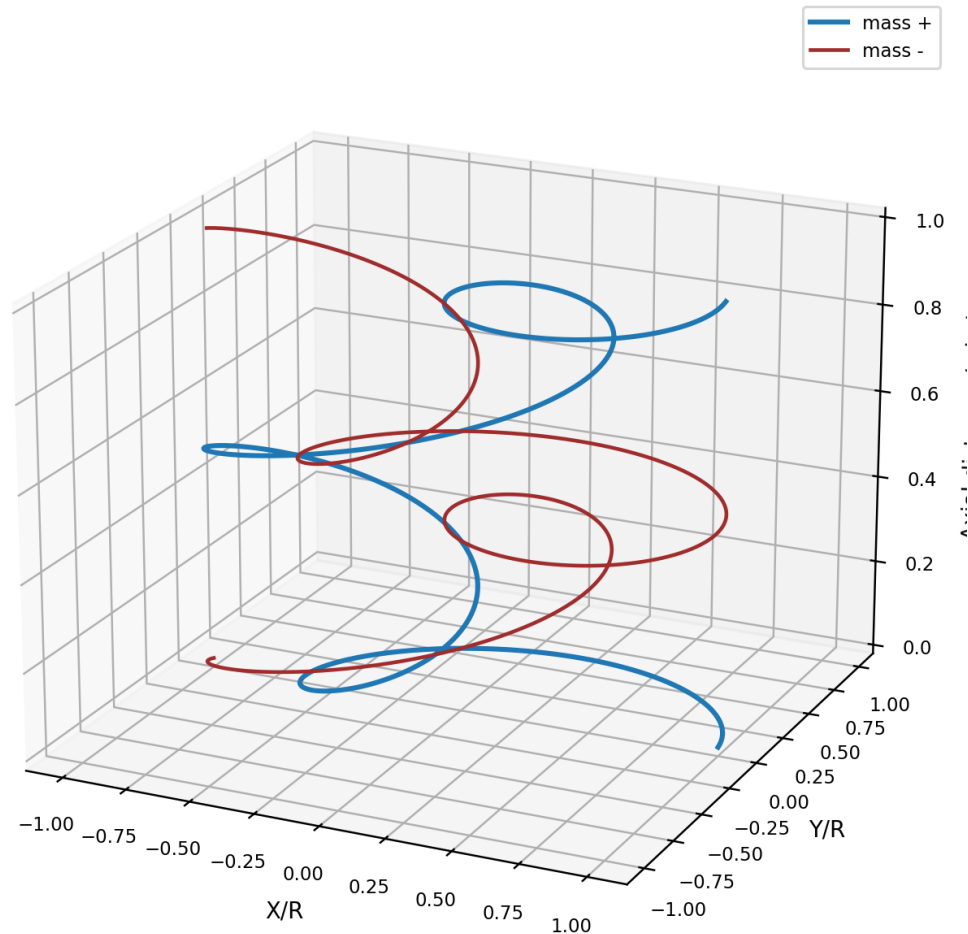


Figure 7. Normalized compound braid. The same two endpoint paths are normalized by rotor radius R and cycle pitch P_B . This dimensionless view reveals the four-lobed two-strand structure hidden by the global scale ratio.

6. Translation periodicity, pitch, and scale separation

The local endpoint direction is periodic with period 2π . The absolute path is therefore not closed but is repeated after a pure translation along the CMB axis:

$$r_{\text{plus/minus}}(\phi + 2\pi) = r_{\text{plus/minus}}(\phi) + P_B \hat{c}$$

$$P_B = 2\pi V_{\text{CMB}} / \omega_B$$

For $\omega_B = 1$ rad/s, the period is 6.283185307180 s and the pitch is 2323600.478980275 m = 2323.600478980 km. The pitch-to-bar-diameter ratio is 2323600.478980. Thus the exact global path is geometrically a fine bounded modulation around an almost straight CMB-directed centerline.

The term pitch is used in the translation-periodic sense. The path is not a circular helix because its transverse radius is not constant and its axial coordinate contains the oscillatory term $\pm R \sin(2\phi)$. A precise description is an axially translated spherical Lissajous curve, or equivalently a four-lobed compound braid with one outer winding and two axial oscillations per translation period.

6.1 Important scale laws

Quantity	Exact dependence
Outer-cycle duration	$2\pi / \omega_B$
CMB pitch per cycle	$2\pi V_{CMB} / \omega_B$
Maximum transverse radius	R
Mass separation	2R at every phase
Local curve shape	Independent of ω_B
Absolute pitch	Inversely proportional to ω_B
Cell-plane crossing rate in time	Proportional to V_{CMB} / ℓ_K

Changing ω_B changes how much global translation occurs during one mechanical cycle, but it does not change the local spherical locus. Increasing rotation rate compresses the repeated local pattern along the CMB centerline. Decreasing rotation rate stretches it.

7. Cell3 base geometry and boundary-plane system

The K-Field base geometry is an oblique parallelepiped generated by the three ICRS-embedded basis vectors a, b, and c. The centered primitive cell consists of coefficients from -1/2 to +1/2 along each basis direction. Its 27 half-step nodes use coefficient indices i,j,k in {-1,0,1} divided by two.

Basis	ICRS x	ICRS y	ICRS z	Length
a	0.020381438517	-0.003416414496	-0.002332965917	0.020797058782
b	0.002198745968	0.012902170259	-0.034066501536	0.036494205131
c	-0.000630379485	-0.024257722017	-0.042924354571	0.049308565900

Primitive Cell3 volume: 2.842066937809649e-05 native length cubed.

Three central Cell3 plane families and their ICRS normals

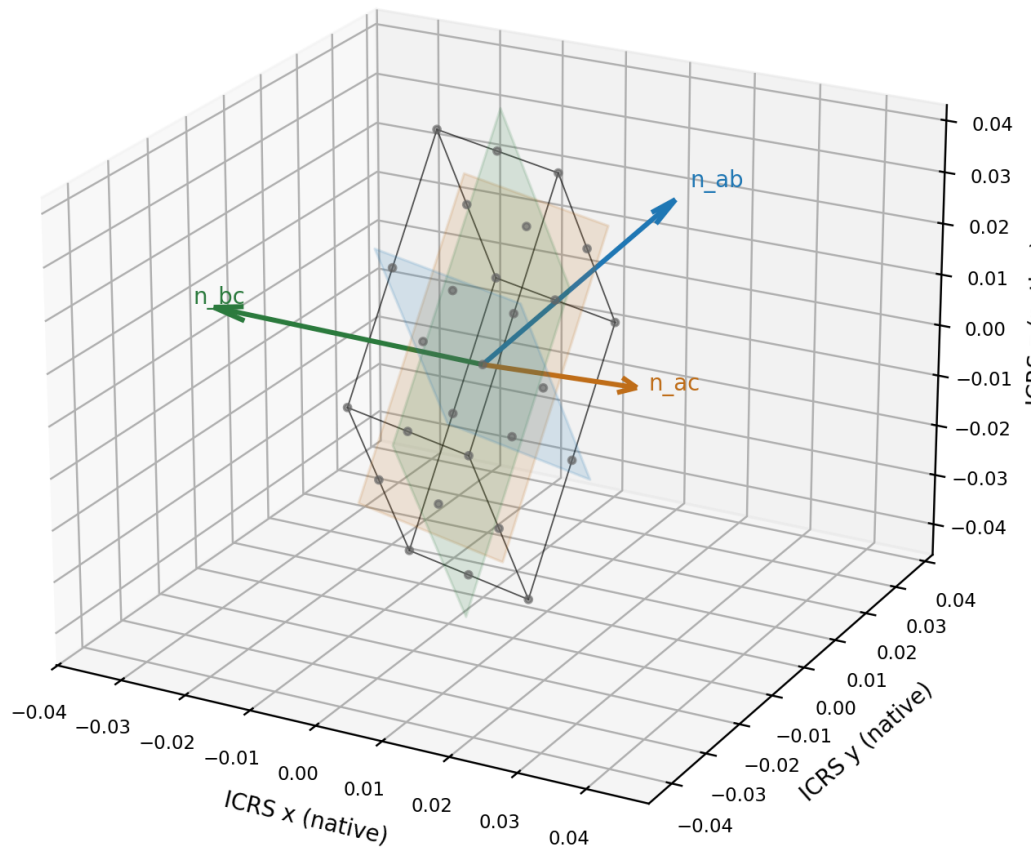


Figure 8. Cell3 plane families. The ab, ac, and bc central planes are drawn with their source normals. Their translated positive and negative members form the six cell boundaries.

7.1 Face-plane equations

Family	Area	Included angle (deg)	Unit normal (x,y,z)	Boundary d
A_ab	7.547217896726e-04	83.933483012	(0.194092201, 0.913177169, 0.358379235)	0.018828573500
A_ac	1.011255757688e-03	80.448130018	(0.089052594, 0.866576769, -0.491033948)	0.014052166903
A_bc	1.385784431733e-03	50.363231814	(-0.995965190, 0.083602163, -0.032619291)	0.010254361619

Every boundary plane has equation $\mathbf{n} \cdot \mathbf{x} = d$. The opposite member of the same family uses $-d$. The face families are named by the two basis vectors that span the face: ab faces have constant c coefficient, ac faces have constant b coefficient, and bc faces have constant a coefficient.

8. Earth-CMB traversal expressed in Cell3 coordinates

Let $\mathbf{B}_K = [\mathbf{a} \ \mathbf{b} \ \mathbf{c}]$ be the Cell3 basis matrix. A native ICRS displacement $\mathbf{x}$ is represented by Cell3 coefficient vector $\mathbf{x}_i = \mathbf{B}_K^{-1} \mathbf{x}$. Applying this map to the CMB unit direction gives:

$$\mathbf{B}_K^{-1} \mathbf{c}_{\text{hat}} = (-48.18103814200, 5.42930495311, 1.12575306624)$$

$$\text{Normalized coefficient direction} = (-0.99344308278, 0.11194664245, 0.02321186176)$$

The dominant negative a coefficient means that the line advances primarily toward decreasing a index. The b and c coefficients increase more slowly. Since none of the components is zero and no pair forms a face-tangent constraint, the CMB line is oblique to

all three face families and to all three primitive edges.

Earth-CMB line through the primitive Cell3 and boundary-plane intersections

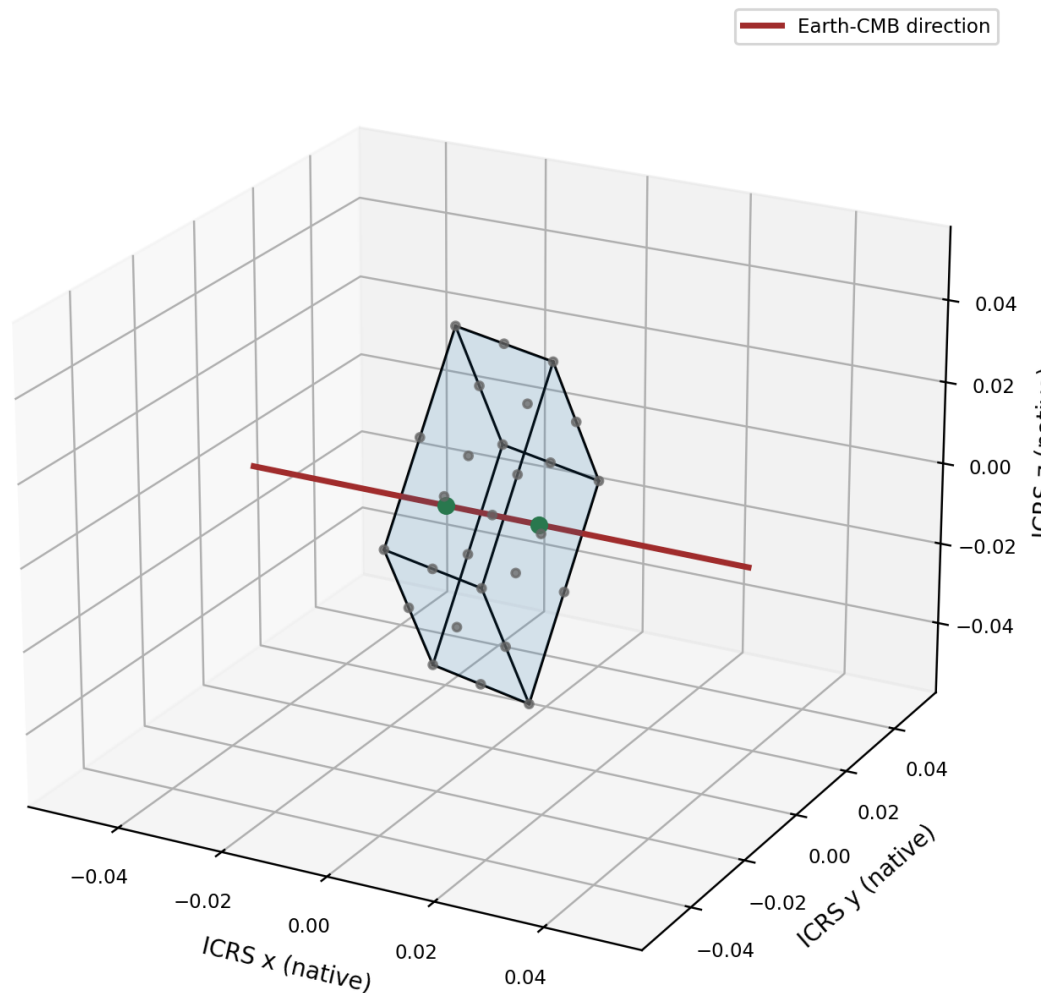


Figure 9. CMB line through the primitive Cell3. The red line is the exact Earth-CMB direction. Colored points identify its intersections with the six boundary planes of the centered primitive cell.

8.1 Seasonal datum in Cell3 coefficients

$$B_K^{-1} \hat{x}_{ICRS} = (48.56300310584, 3.16864276926, -5.15419292932)$$

If one native K-Field length equals ℓ_K metres and the absolute Cell3 origin has coefficient vector $\mathbf{x}_{i_origin}$, the RA=0 Earth-center datum is:

$$\mathbf{x}_{i_E0} = \mathbf{x}_{i_origin} + (AU/\ell_K) B_K^{-1} \hat{x}_{ICRS}$$

The source establishes the native geometry and ICRS orientation but does not assign ℓ_K in metres or the absolute global Cell3 origin. Consequently, the directional crossing sequence is exact, while the absolute cell number and physical boundary spacing remain parameterized by ℓ_K and $\mathbf{x}_{i_origin}$.

8.2 Complete endpoint traversal in Cell3 coefficients

$$\mathbf{x}_{i_plus/minus}(\phi) = \mathbf{x}_{i_E0} + [V_{CMB} \phi / (\omega_B \ell_K)] B_K^{-1} \hat{c}_{hat} \pm [R/\ell_K] B_K^{-1} \mathbf{u}(\phi)$$

This equation separates the Cell3 traversal into a linear drift and a bounded periodic modulation. The drift determines the long-run plane-crossing order. The periodic term advances or delays individual crossings and moves the endpoints laterally among neighboring coefficient directions.

9. Plane-crossing sequence and traversal address

In coefficient space, the boundary planes occur at half-integer coordinates. The bc family is $xi_a=k+1/2$, the ac family is $xi_b=k+1/2$, and the ab family is $xi_c=k+1/2$. For the centerline $x=\tau c_{\hat{}}$, the coefficient path is $xi=\tau d$ with $d=B_K^{-1} c_{\hat{}}$. Consecutive crossings of one family are separated by $1/|d_i|$ in native ICRS distance τ .

Plane family	Coefficient condition	Direction component	Centerline spacing (native length)	Crossing density
bc / constant a	$xi_a=k+1/2$	-48.18103814200	0.02075505300	48.18103814200
ac / constant b	$xi_b=k+1/2$	5.42930495311	0.18418563861	5.42930495311
ab / constant c	$xi_c=k+1/2$	1.12575306624	0.88829427162	1.12575306624

Exact Cell3 boundary-plane crossing sequence over one native length

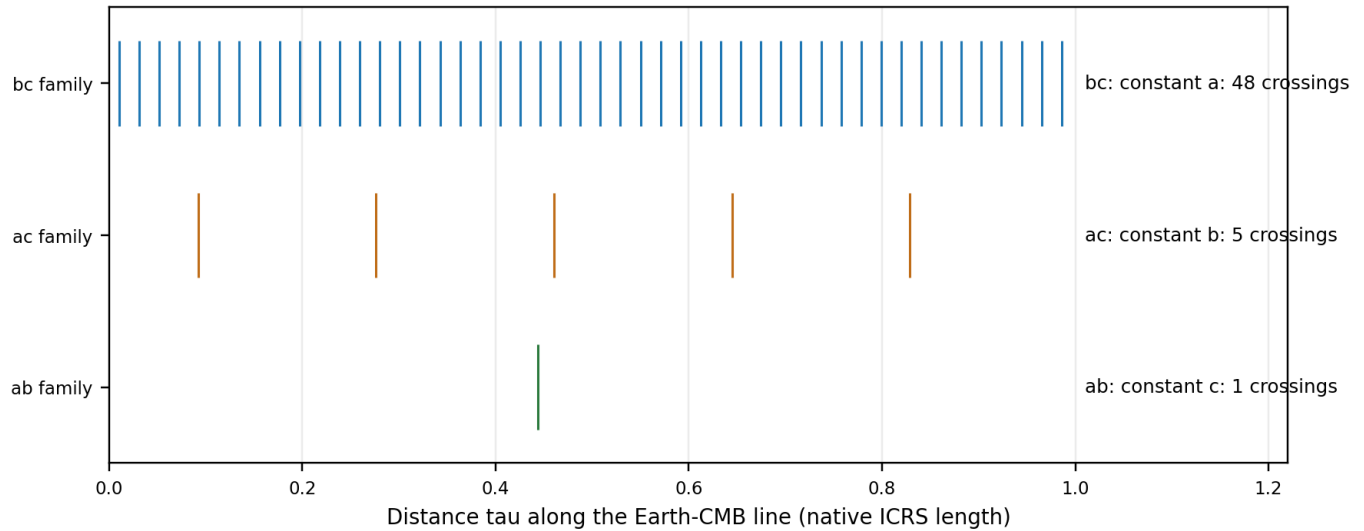


Figure 10. Boundary crossing sequence. Each vertical mark is an exact half-integer plane crossing of the Earth-CMB centerline over one native ICRS length. The bc family dominates the sequence.

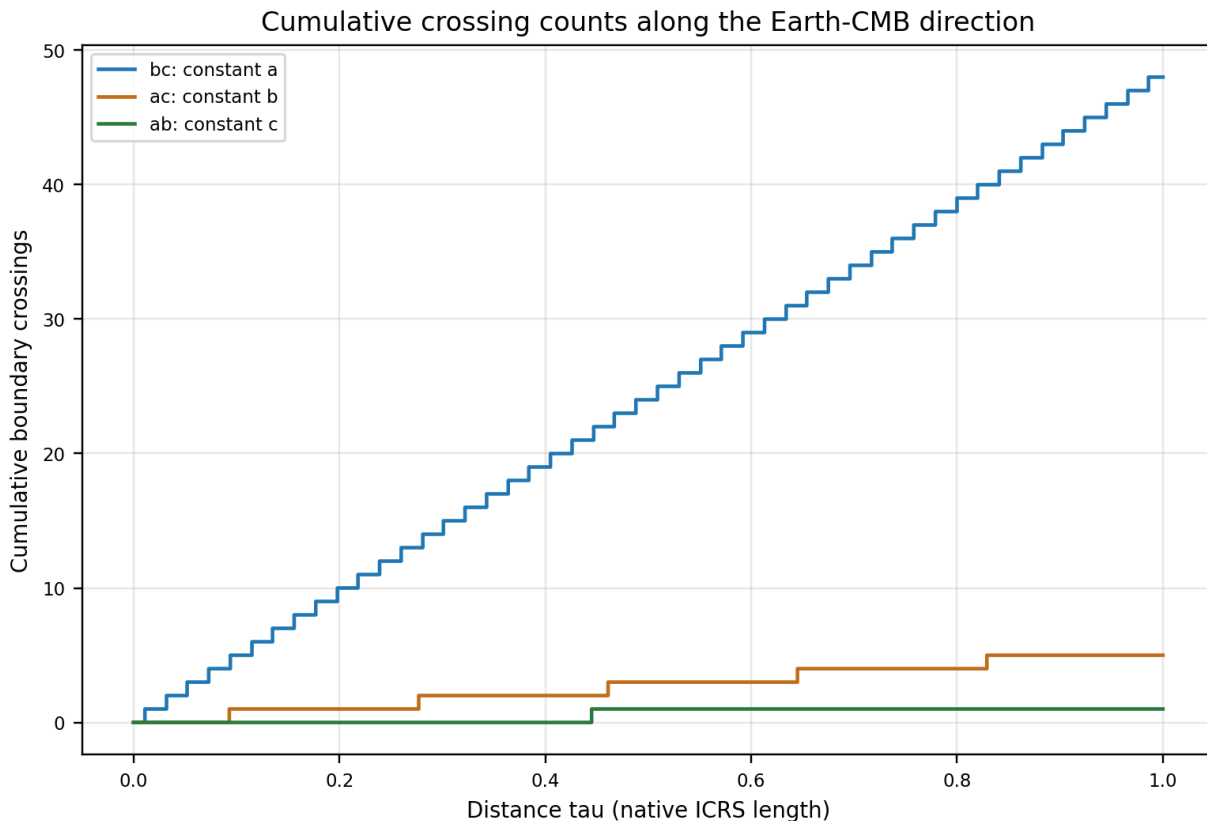


Figure 11. Cumulative plane crossings. The staircase counts demonstrate the asymptotic crossing densities
48.18103814:5.42930495:1.12575307.

The crossing stream can be encoded as an ordered word over three symbols: bc, ac, and ab. Adding the integer plane index yields a complete centerline boundary address. Adding the rotor phase and the plus/minus endpoint branch identifies the exact local position relative to the centerline. This creates a deterministic mapping from mechanical phase to K-Field traversal state.

9.1 Physical spacing with an explicit scale bridge

$$\text{Delta_s_bc} = \text{ell_K} / 48.18103814$$

$$\text{Delta_s_ac} = \text{ell_K} / 5.42930495$$

$$\text{Delta_s_ab} = \text{ell_K} / 1.12575307$$

If ell_K is later calibrated in metres, these formulas immediately convert the native crossing sequence into physical distance and time. At Earth-CMB speed, the corresponding mean crossing frequencies are $V_{\text{CMB}}/\text{Delta_s}$ for each family.

10. Relation to the Q-metric structure

The path geometry is defined by the two rotations, the Earth-CMB direction, and the Cell3 basis. The Q matrix does not alter the path itself; it assigns directional phase-transition cost and kinetic weighting to vectors along that path. Separating geometry from metric weighting is essential: the same curve can be evaluated under Euclidean length, Q-metric energy, or plane-crossing address without changing its ICRS position.

$$C(n) = n^T Q n$$

$$E_Q(v) = (1/2) m \kappa_F v^T Q v$$

Unit-level Q-metric ellipsoid with Earth-CMB traversal axis

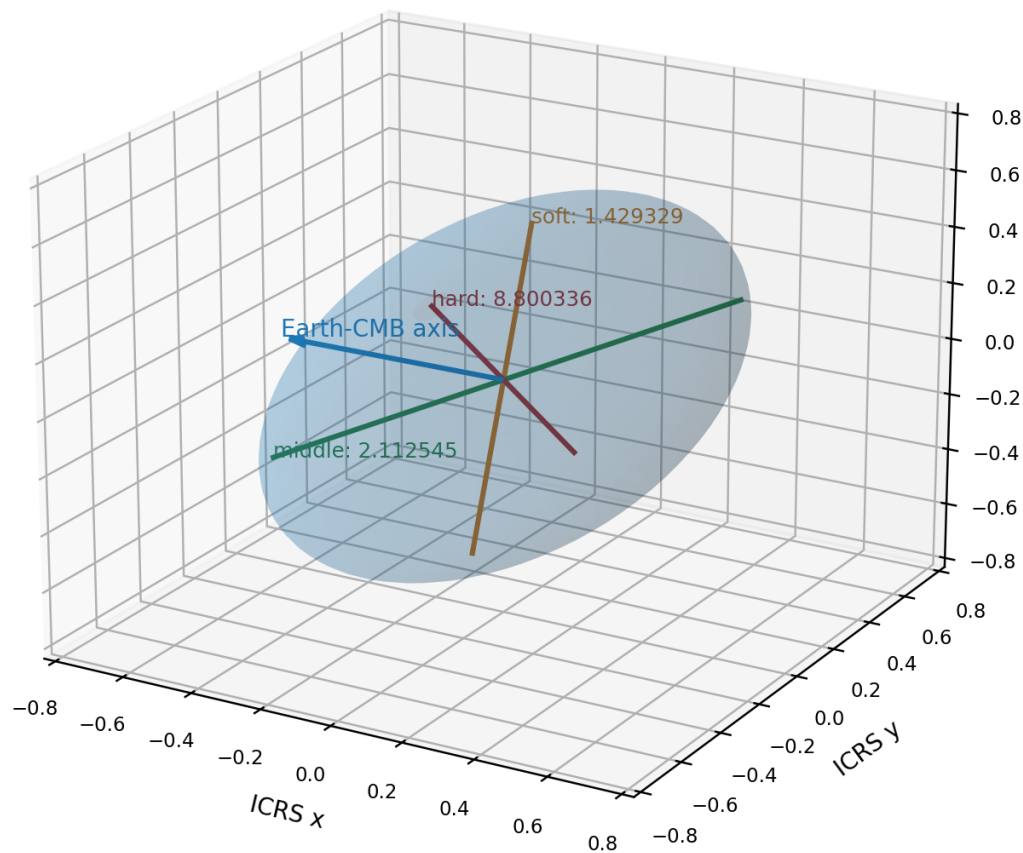


Figure 12. Q-metric ellipsoid. The Earth-CMB traversal axis is drawn through the unit-level ellipsoid $x^T Q x = 1$. The ellipsoid shows the anisotropic directional cost encountered by the moving endpoints.

As the mass direction $u(\phi)$ and tangent $du/d\phi$ evolve, they sample different combinations of the soft, middle, and hard principal directions of Q . Translation by V_E adds a common global velocity. The source kinetic analysis evaluates the resulting cross term $u_{\phi}^T Q V_E$. This reference preserves that metric context while isolating the exact geometric path on which the metric is evaluated.

11. Geometric interpretation of one complete traversal cycle

At $\phi=0$, mass plus is displaced by $+R e_0$ from Earth center and mass minus by $-R e_0$. The Earth center then advances along $c_{\hat{a}}$. During the first quarter of the outer cycle, the transverse radius contracts and mass plus approaches the positive CMB-axis side of the local sphere. At $\phi=\pi/4$ its transverse displacement is zero and its local axial offset is $+R$. Mass minus is simultaneously on the same centerline at local axial offset $-R$.

From $\pi/4$ to $3\pi/4$ the pair expands away from the centerline, crosses a transverse equatorial direction, and contracts again. At $\phi=3\pi/4$ mass plus reaches local axial offset $-R$. The sequence repeats in the second half-cycle with the transverse direction reversed by the outer rotation. Four centerline passages and four expanded lobes occur during one outer period.

While this local cycle unfolds, Earth center moves 2323.600479 km. The local pole contacts that coincide on the stationary radius- R sphere are therefore separated by hundreds of kilometres along the fixed K-Field trajectory. The global object is not a

closed loop but a repeated spatial motif whose phase is tied to position along the CMB line.

11.1 Canonical geometric classification

LOCAL CLASSIFICATION: a closed immersed spherical algebraic curve with 1:2 rotational winding and 1:2:3 harmonic content.

GLOBAL CLASSIFICATION: an open translation-periodic spherical trochoidal braid confined to a radius-R tube around the Earth-CMB axis.

CELL3 CLASSIFICATION: an oblique line-plus-periodic-oscillation traversal that intersects all three half-integer face-plane stacks in a deterministic ordered sequence.

The concise canonical description is:

A translationally periodic, antipodal, four-lobed compound braid advancing along the Earth-CMB vector and cutting obliquely through all three Cell3 plane stacks.

12. Educational consequences and discovery pathways

The construction creates a direct bridge among mechanical phase, celestial orientation, and discrete K-Field structure. A phase value ϕ determines the local mass direction, tangent, axial position, transverse radius, absolute ICRS position, Cell3 coefficient position, and the next plane-family crossing. These are not separate descriptions; they are coordinate representations of one geometric event.

This changes the operative analytical paradigm within the K-Field program. A rotating device is not treated only as a closed motion in a laboratory frame. It is treated as a directed traversal through an anisotropic, plane-structured spatial fabric. The local cycle becomes a repeated modulation of a much longer CMB-directed path. Seasonal location fixes the absolute entry datum, while Cell3 orientation fixes the sequence of structural boundaries encountered.

The geometry supports several concrete investigations: phase-resolved comparison of plane crossings with measured force or energy channels; seasonal changes produced by changing the Earth-center datum while retaining the same CMB axis; sensitivity to alternate inner-to-outer rate ratios; identification of rate locks that produce closed local curves and simple translation periods; and extension from point masses to distributed rotor geometries by integrating the same path map over material phase.

No additional force law is required to define these paths. Force, energy, or signal predictions are secondary functions evaluated on the geometry. This separation permits geometry to be audited independently before constitutive models are applied.

13. Reproduction procedure

Step	Operation
1	Load the source ICRS vectors $\hat{c}$, $\hat{e}_0$, and A_0 ; verify that they form a right-handed orthonormal triad.
2	Set $R=0.5$ m, $\omega_A=2\omega_B$, and $\phi=\omega_B t$.
3	Construct $\hat{e}(\phi)=\cos(\phi)\hat{e}_0+\sin(\phi)A_0$ and $\hat{u}(\phi)=\cos(2\phi)\hat{e}(\phi)+\sin(2\phi)\hat{c}$.
4	Construct the local endpoint positions $\pm R\hat{u}(\phi)$.
5	Select the seasonal datum $r_{E0}=AU(1,0,0)$ for $RA=0$ deg, $Dec=0$ deg.
6	Construct Earth-center translation $(V_{CMB}/\omega_B)\hat{c}$.
7	Add the local endpoint positions to the translated Earth-center position.
8	Load the Cell3 basis matrix $B_K=[a\ b\ c]$ and compute $B_K^{-1}\hat{c}$ and $B_K^{-1}\hat{x}$.
9	Introduce ell_K only when converting metres into native K-Field length; do not silently equate the two units.
10	Locate boundary crossings from half-integer coefficient conditions $x_i{}_a, x_i{}_b, x_i{}_c = k+1/2$.
11	Check translation periodicity after ϕ increases by 2π .
12	Compare numerical path samples with the formulas and the machine-readable file accompanying this report.

14. Canonical result statement

For an Earth-center dual-axis device with endpoint radius R , outer axis $\hat{c}$ aligned to the Earth-CMB vector, perpendicular rotating transverse reference $\hat{e}(\phi)$, and exact inner-to-outer rate ratio 2:1, the local positive endpoint direction is:

$$u(\phi) = \cos(2\phi) [\cos(\phi)\hat{e}_0 + \sin(\phi)\hat{A}_0] + \sin(2\phi)\hat{c}$$

With the $RA=0$ deg seasonal datum $r_{E0} = AU \hat{x}_{ICRS}$ and constant CMB translation, the absolute endpoint paths are:

$$r_{\pm}(\phi) = r_{E0} + (V_{CMB}/\omega_B)\phi \hat{c} \pm R u(\phi)$$

They obey the translation-periodicity law:

$$r_{\pm}(\phi + 2\pi) = r_{\pm}(\phi) + (2\pi V_{CMB}/\omega_B)\hat{c}$$

At $\omega_B = 1$ rad/s and $R = 0.5$ m, the local curve length is 6.659167222 m, the cycle duration is 6.283185307 s, and the translation pitch is 2323.600478980 km. In Cell3 coefficients the Earth-CMB direction is (-48.18103814200, 5.42930495311, 1.12575306624), producing exact plane-crossing densities (48.18103814200, 5.42930495311, 1.12575306624) for the bc, ac, and ab families, respectively.

The resulting global geometry is a translationally periodic, antipodal, four-lobed compound braid cutting obliquely through all three Cell3 plane stacks along the Earth-CMB vector.

Appendix A. Complete Cell3 boundary-vertex table

#	i	j	k	ICRS x	ICRS y	ICRS z
0	-0.5	-0.5	-0.5	-0.010974902500	0.007385983127	0.039661911012
1	-0.5	-0.5	0.5	-0.011605281985	-0.016871738890	-0.003262443559
2	-0.5	0.5	-0.5	-0.008776156532	0.020288153386	0.005595409476
3	-0.5	0.5	0.5	-0.009406536017	-0.003969568631	-0.037328945095
4	0.5	-0.5	-0.5	0.009406536017	0.003969568631	0.037328945095
5	0.5	-0.5	0.5	0.008776156532	-0.020288153386	-0.005595409476
6	0.5	0.5	-0.5	0.011605281985	0.016871738890	0.003262443559
7	0.5	0.5	0.5	0.010974902500	-0.007385983127	-0.039661911012

Appendix B. Complete 27-node centered half-step registry

#	2i	2j	2k	lambda	ICRS x	ICRS y	ICRS z
0	-1	-1	-1	-3	-0.010974902500	0.007385983127	0.039661911012
1	-1	-1	0	-2	-0.011290092243	-0.004742877881	0.018199733726
2	-1	-1	1	-1	-0.011605281985	-0.016871738890	-0.003262443559
3	-1	0	-1	-2	-0.009875529516	0.013837068257	0.022628660244
4	-1	0	0	-1	-0.010190719258	0.001708207248	0.001166482958
5	-1	0	1	0	-0.010505909001	-0.010420653760	-0.020295694327
6	-1	1	-1	-1	-0.008776156532	0.020288153386	0.005595409476
7	-1	1	0	0	-0.009091346274	0.008159292378	-0.015866767810
8	-1	1	1	1	-0.009406536017	-0.003969568631	-0.037328945095
9	0	-1	-1	-2	-0.000784183241	0.005677775879	0.038495428054
10	0	-1	0	-1	-0.001099372984	-0.006451085129	0.017033250768
11	0	-1	1	0	-0.001414562727	-0.018579946138	-0.004428926518
12	0	0	-1	-1	0.000315189743	0.012128861009	0.021462177286
13	0	0	0	0	0.000000000000	0.000000000000	0.000000000000
14	0	0	1	1	-0.000315189743	-0.012128861009	-0.021462177286
15	0	1	-1	0	0.001414562727	0.018579946138	0.004428926518
16	0	1	0	1	0.001099372984	0.006451085129	-0.017033250768
17	0	1	1	2	0.000784183241	-0.005677775879	-0.038495428054
18	1	-1	-1	-1	0.009406536017	0.003969568631	0.037328945095
19	1	-1	0	0	0.009091346274	-0.008159292378	0.015866767810
20	1	-1	1	1	0.008776156532	-0.020288153386	-0.005595409476
21	1	0	-1	0	0.010505909001	0.010420653760	0.020295694327
22	1	0	0	1	0.010190719258	-0.001708207248	-0.001166482958
23	1	0	1	2	0.009875529516	-0.013837068257	-0.022628660244
24	1	1	-1	1	0.011605281985	0.016871738890	0.003262443559
25	1	1	0	2	0.011290092243	0.004742877881	-0.018199733726
26	1	1	1	3	0.010974902500	-0.007385983127	-0.039661911012

Appendix C. Representative one-cycle traversal samples

The machine-readable companion contains all 361 one-degree samples. The table below gives 15-degree witness samples in the path frame and the fixed K-Field translation coordinate. Axial distance is measured from the RA=0 datum along c_hat; transverse coordinates are measured in metres along e0 and A0.

phi (deg)	inner (deg)	time (s)	X+ (m)	Y+ (m)	Z+ local (m)	s+ (km)	rho (m)
0	0	0.000000000	0.500000000	0.000000000	0.000000000	0.000000000	0.500000000
15	30	0.261799388	0.418258152	0.112071934	0.250000000	96.816936624	0.433012702
30	60	0.523598776	0.216506351	0.125000000	0.433012702	193.633806261	0.250000000
45	90	0.785398163	0.000000000	0.000000000	0.500000000	290.450559873	0.000000000
60	120	1.047197551	-0.125000000	-0.216506351	0.433012702	387.267179509	0.250000000
75	150	1.308996939	-0.112071934	-0.418258152	0.250000000	484.083683121	0.433012702
90	180	1.570796327	0.000000000	-0.500000000	0.000000000	580.900119745	0.500000000
105	210	1.832595715	0.112071934	-0.418258152	-0.250000000	677.716556369	0.433012702
120	240	2.094395102	0.125000000	-0.216506351	-0.433012702	774.533059981	0.250000000
135	270	2.356194490	0.000000000	-0.000000000	-0.500000000	871.349679618	0.000000000
150	300	2.617993878	-0.216506351	0.125000000	-0.433012702	968.166433229	0.250000000
165	330	2.879793266	-0.418258152	0.112071934	-0.250000000	1064.983302866	0.433012702
180	360	3.141592654	-0.500000000	0.000000000	-0.000000000	1161.800239490	0.500000000
195	390	3.403392041	-0.418258152	-0.112071934	0.250000000	1258.617176114	0.433012702
210	420	3.665191429	-0.216506351	-0.125000000	0.433012702	1355.434045751	0.250000000
225	450	3.926990817	-0.000000000	-0.000000000	0.500000000	1452.250799363	0.000000000
240	480	4.188790205	0.125000000	0.216506351	0.433012702	1549.067419000	0.250000000
255	510	4.450589593	0.112071934	0.418258152	0.250000000	1645.883922611	0.433012702
270	540	4.712388980	0.000000000	0.500000000	0.000000000	1742.700359235	0.500000000
285	570	4.974188368	-0.112071934	0.418258152	-0.250000000	1839.516795859	0.433012702
300	600	5.235987756	-0.125000000	0.216506351	-0.433012702	1936.333299471	0.250000000
315	630	5.497787144	-0.000000000	0.000000000	-0.500000000	2033.149919108	0.000000000
330	660	5.759586532	0.216506351	-0.125000000	-0.433012702	2129.966672719	0.250000000
345	690	6.021385919	0.418258152	-0.112071934	-0.250000000	2226.783542356	0.433012702
360	720	6.283185307	0.500000000	-0.000000000	-0.000000000	2323.600478980	0.500000000

Appendix D. Formula and variable ledger

Symbol	Meaning	Units or type
B_K	Cell3 basis matrix [a b c]	native ICRS length
c_hat	Earth-CMB unit direction and axis B	unit vector
V_E	Earth-CMB velocity vector	m/s
V_CMB	Magnitude of V_E	m/s
e0	Bar reference at phi=0	unit vector
A0	Axis A reference at phi=0	unit vector
phi	Outer phase omega_B t	radian
u(phi)	Positive endpoint direction	unit vector
R	Mass radius from Earth center	m
omega_B	Outer angular rate	rad/s
omega_A	Inner angular rate, 2 omega_B	rad/s
r_E0	RA=0 Earth-center seasonal datum	m, ICRS vector
P_B	Translation pitch per outer cycle	m
ell_K	Physical length corresponding to one native K-Field length	m/native length
xi	Cell3 coefficient coordinate	dimensionless coefficient
Q	Quadratic phase-transition metric	3x3 SPD matrix

Appendix E. Source and computational ledger

This report is computed from the attached K-Field dual-axis kinetic mass analysis PDF and JSON, archival epoch 1784741897. The title-page and frontispiece hierarchy follows the established Krytronx archival report format represented in the historical document library. Numerical figures are generated directly from the source vectors, Cell3 basis, nodes, planes, Q matrix, rotor radius, and Earth-CMB velocity.

Item	Value
Source PDF	K-Field_dual_axis_kinetic_mass_analysis_1784741897(1).pdf
Source JSON	K-Field_dual_axis_kinetic_mass_analysis_1784741897(1).json
Report epoch	1784745791
Machine-readable companion	K-Field_dual_axis_kinetic_mass_CMB_traversal_geometry_1784745791.json
Figure sampling	721 phase points over 0 to 2pi
Tabular path sampling	361 one-degree samples
ReportLab fonts	Helvetica, Helvetica-Bold, Times-Roman, Courier
Page size	US Letter
Margins	0.68 inch on all sides
Cell wrapping	Every table text cell is a ReportLab Paragraph
Long-table headers	repeatRows=1

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